

Tushar Ghosh

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EDUCATION

CHRIST (Deemed to be University)

Master of Computer Applications, 87%

Bangalore, Karnataka

June 2024 – May 2026

REVA University

Bachelor of Computer Applications, 82%

Bangalore, Karnataka

July 2020 – October 2023

EXPERIENCE

Software Developer

CISOGenie

June 2025 – June 2026

- Revamped and created enterprise dashboard across 8+ frontend modules, improving usability and introducing modern UI patterns such as bento grid layouts for better clarity and structured information flow.
- Built a compliance readiness application supporting 3 frameworks (ISO 27001, SOC 2, and related standards), enabling assessment workflows, report generation, PDF export, and white-labeled enterprise outputs.
- Developed a Business Impact Analysis (BIA) module handling system, people, and process entities with risk mapping, enabling dynamic BIA generation and real-time updates via structured tracking tables.
- Collaborated in Agile teams using Git, GitHub, and GitLab to deliver production-grade frontend features across multiple enterprise releases.

Web & Frontend Engineer (Freelance)

APLYD (Athena Infonomics)

July-August 2026

- Architected and developed a high-performance static website using **Astro and TypeScript**, leveraging Island Architecture across **19+ routes** to minimize client-side JavaScript and deliver fast, responsive page loads.
- Engineered interactive data visualizations and motion systems, including a **Canvas-based global footprint map**, ambient particle animations, responsive SVG workflow diagrams, and branded micro-interactions with **prefers-reduced-motion** support.
- Built modular **multi-sector filtering and taxonomy systems** enabling seamless navigation across 19+ case studies spanning government, nonprofits, philanthropy, agriculture, education, utilities, and civic services.
- Implemented an enterprise-grade **technical SEO and Schema.org JSON-LD @graph** system with dynamic breadcrumbs, canonical URLs, XML sitemaps, OpenGraph/Twitter cards, and interconnected Organization, WebSite, Service, FAQPage, and ItemList entities.

PROJECTS

ArchitectAI | Python, FastAPI, LangGraph, LangChain, RAG, ChromaDB, NetworkX, React, TypeScript

- Built an AI-powered codebase intelligence platform that analyzes uploaded repositories through static analysis, dependency graph generation, and retrieval-augmented generation to provide architecture insights and conversational exploration.
- Implemented AST and tree-sitter based parsing pipelines to extract real import relationships, generate NetworkX dependency graphs, and create Mermaid architecture diagrams grounded in repository structure.
- Designed a LangGraph tool-calling agent with ChromaDB vector retrieval and local embeddings to answer codebase questions using verified source context instead of LLM-only reasoning.
- Developed a React + TypeScript interface with Mermaid rendering for interactive dependency visualization, architecture diagrams, and AI-assisted repository analysis.

MRanalysis | Python, FastAPI, PostgreSQL, Redis, Celery, Docker, React, TypeScript

- Engineered a production-ready backend that clones public Git repositories, traverses complete Git history, and computes repository analytics including contributor activity, code ownership, commit trends, file growth, and bus-factor metrics without relying on the GitHub API.
- Designed an asynchronous processing pipeline using FastAPI, Celery, Redis, and PostgreSQL where repository analyses execute as background jobs while clients track progress through polling and receive precomputed analytics through REST APIs.

- Implemented JWT authentication, report ownership, and persistent storage of repository analysis snapshots, enabling secure access to previously generated reports.
- Developed a React + TypeScript dashboard for interactive visualization of repository analytics and deployed the frontend on Vercel and backend on Render using Docker for production-ready delivery.

Signwise | *Next.js 16, FastAPI, Google Gemini, Pydantic v2, PyPDF, Python-docx*

- Architected an end-to-end legal intelligence platform that translates complex agreements into plain English, performs AI-powered risk audits, and generates counter-proposals using Next.js 16, FastAPI, and Google Gemini.
- Engineered an in-memory document parsing pipeline using PyPDF and Python-docx to securely process files up to 25MB, sanitize byte streams, and extract structured clause chunks while preserving document metadata.
- Implemented a resilient AI pipeline with strict Pydantic v2 JSON schemas, automatic model fallback, and retry logic, ensuring type-safe serialization between backend APIs and frontend TypeScript clients.
- Built a high-trust legal dashboard featuring bilateral obligation mapping, surrendered-rights matrices, section-grounded QA, and one-click HR negotiation email generation.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript

Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS, React Router, TanStack Query, Mermaid

Backend: FastAPI, Node.js, REST APIs, JWT Authentication, Celery, Uvicorn

AI / LLM Engineering: LangGraph, LangChain, Retrieval-Augmented Generation (RAG), ChromaDB, Sentence Transformers, Groq LLMs

Code Analysis & Graph Processing: Python AST, Tree-sitter, NetworkX, Dependency Graph Analysis

Databases & Storage: PostgreSQL, MongoDB, Firebase (Auth, Firestore), Redis

Cloud & DevOps: Docker, AWS EC2, Render, Vercel

Developer Tools: Git, GitHub, GitLab, Postman, VS Code

Concepts: System Design, Asynchronous Processing, Repository Analytics, Agentic Workflows, Responsive Design, Agile Development, CI/CD Basics